

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses conservation of energy to form an equation.	AO3.4	M1	$\frac{1}{2}mv^2 = mg\left(\frac{r}{4} + (r - r\cos\theta)\right)$
	Forms correct equation.	AO1.1b	A1	$\frac{1}{2}v^2 = gr\left(\frac{1}{4} + 1 - \cos\theta\right)$
	Completes mathematical argument to find v. AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips with the change in height shown clearly and correctly.	AO2.1	R1	$v^2 = 2gr\left(\frac{5}{4} - \cos\theta\right)$ $= \frac{gr}{2}(5 - 4\cos\theta)$ $v = \sqrt{\frac{gr}{2}(5 - 4\cos\theta)}$ AG
(b)	Resolves radially to form equation to find θ .	AO3.4	M1	$mg\cos\theta = \frac{m}{r}\frac{gr}{2}(5 - 4\cos\theta)$
	Obtains correct value for θ from 'their' equation FT provided both M1 marks awarded	AO1.b	A1F	$2\cos\theta = 5 - 4\cos\theta$ $\cos\theta = \frac{5}{6}$ $\theta = 34^\circ$
(c)	Explains that air resistance will decrease the speed	AO3.5a	E1	Air resistance would decrease the speed and therefore increase θ
	Deduces that θ will increase	AO2.2a	R1	
(d)	Explains that the normal reaction will change as the child moves down the slope and friction is proportional to the normal reaction	AO2.4	E1	Because $F = \mu N$ and N varies as child moves down the slope
				Total 8 marks

Q2.

- (a) Using conservation of energy:

$$\frac{1}{2} mv^2 = \frac{1}{2} mu^2 - mgh$$

for 3 terms, 2 KE and 1 PE

M1

$$\frac{1}{2} \times 3 \times v^2 = \frac{1}{2} \times 3 \times 4^2 - 3 \times g \times 1.2(1 - \cos 25)$$

M1A1 for finding h [M1 for 1.2(1 - cos 25 or sin 25)]

M1

A1

$$v^2 = 4^2 - 2.4 \times g (1 - \cos 25)$$

$$v^2 = 16 - 2.2036$$

$$v = 3.71 \text{ m s}^{-1}$$

Accept 3.7, 3.70, 3.72

A1

4

(b) Resolving radially:

$$T = mg \cos 25 + \frac{mv^2}{r}$$

M1 accept cos 25 or sin 25, + or - sign and ≠ 2

$$= 26.645 + 34.491$$

A1 fully correct and substituted

M1A1

$$= 61.1 \text{ N}$$

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Accept 61.0 or 61

A1

3

[7]

Q3.

(a) Using conservation of energy:

$$\frac{1}{2} m(5u)^2 = \frac{1}{2} m(2u)^2 + 2amg$$

M1 for 3 [or 4] terms: 2 KE
and 1[or 2] PE

M1A1

$$\frac{1}{2} \times 21 \times u^2 = 2ag$$

M1A1 for finding h

M1

$$u = \sqrt{\frac{4ag}{21}}$$

A1

4

- (b) Using conservation of energy with speed at point S to be V :

$$\frac{1}{2} m(5u)^2 = \frac{1}{2} m(V)^2 + amg(1 + \cos 60)$$

Or

$$\frac{1}{2} m(V)^2 = amg(1 - \cos 60^\circ) + \frac{1}{2} m \left(2\sqrt{\frac{4ag}{21}} \right)^2$$

$$\frac{1}{2} mV^2 = \frac{1}{2} m(5u)^2 - \frac{1}{2} amg$$

$$V^2 = 25 \times \left(\frac{4ag}{21} \right) - 3ag$$

M1

$$V^2 = \frac{37ag}{21}$$

A1

Resolving radially at point S :

$$R = -mg \cos 60 + \frac{m(V)^2}{a}$$

$$= -\frac{1}{2} mg + \frac{37mg}{21}$$

M1A1

$$= \frac{53}{42} mg \quad \text{or} \quad 1.26mg$$

A1

5

[9]

Q4.

- (a) by conservation of energy:

$$\frac{1}{2}m(u)^2 = \frac{1}{2}m(v)^2 + mg2a$$

*M1 for 3 terms, 2 KE and PE; not
 $v^2 = u^2 + 2as$*

M1

$$v^2 = u^2 - 4ag$$

A1

2

- (b) (i) at point A; $T_1 = \frac{m(v)^2}{a} - mg$
*both signs incorrect M1
either correct M1A1*

M1A1

at point B; $T_2 = \frac{m(u)^2}{a} + mg$
*both signs incorrect M1
either correct M1A1*

A1

$$\frac{T_1}{T_2} = \frac{2}{5}$$

*or $5T_A = 2T_B$
or $T_1 = 2T, T_2 = 5T$*

B1

$$s \left(\frac{m(v)^2}{a} - mg \right) = 2 \left(\frac{m(u)^2}{a} + mg \right)$$

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$$s \left(\frac{m(u^2 - 4ag)}{a} - mg \right)$$

$$= 2 \left(\frac{m(u)^2}{a} + mg \right)$$

CAO

A1

$$5u^2 - 20ag - 5ag = 2u^2 + 2ag$$

from ratio 2 : 5 or 5 : 2 and one tension equation correct

m1

$$3u^2 = 27ag$$

$$u = 3\sqrt{ag}$$

condone $\sqrt{9ag}$

A1

7

(ii) $u^2 = v^2 + 4ag \rightarrow v = \sqrt{5ag}$

condone $v^2 = 5ag$

B1

ratio $u : v = 3 : \sqrt{5}$

accept 1.34 : 1 or 1 : 0.745

B1

2

[11]

Q5.

(a) Using conservation of energy:

$$\frac{1}{2}mv^2 = mgh$$

M1 for 2 or 3 terms, 1 KE and 1 or 2 PE

M1

$$\frac{1}{2}mv^2 = mg2.4(1 - \cos 18)$$

$$v^2 = 4.8g(1 - \cos 18)$$

$$= 2.302$$

m1A1 for finding h

©2025 Exam Papers Practice. All rights reserved. m1A1

$$v = 1.52 \text{ m s}^{-1}$$

Condone 1.51

A1

4

(b) Resolving vertically:

$$T = mg + \frac{mv^2}{r}$$

Correct 3 terms

M1

Correct signs

A1

$$= 22g + \frac{22 \times 2.302}{2.4}$$

$$= 236.7 \dots \text{ N}$$

$$= 237 \text{ N}$$

Accept 236 N

A1

3

[7]

Q6.

- (a) By conservation of energy

$$\frac{1}{2}m(5v)^2 = \frac{1}{2}m(3v)^2 + mg2a$$

M1: for 3 terms, 2 KE and PE

M1

A1

$$8v^2 = 2ag$$

A1

$$v = \sqrt{\frac{ag}{4}} \text{ or } \frac{1}{2}\sqrt{ag}$$

A1

4

- (b) Greatest and least values of tension are at the highest and lowest points of its path

$$\text{At top, } T = \frac{m(3v)^2}{a} - mg$$

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M1

$$= \frac{5}{4}mg$$

ft – must be positive tension

A1ft

$$\text{At B, } T = \frac{m(5v)^2}{a} + mg$$

M1

$$= \frac{29}{4}mg$$

A1ft

Ratio is 29 : 5



A1

5

[9]

Q7.

- (a) Using conservation of energy
(lowest and highest points)

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mg(2a)$$

M1: Equation for conservation of energy with two KE terms and one or two PE terms. May see m or 0.3 .

A1: Correct equation.

M1A1

$$u^2 = v^2 + 4ag$$

For complete revolutions, $v > 0$

$$\therefore u^2 > 4ag$$

$$u > 2\sqrt{ag}$$

AG

A1: Correct result with statement of $v > 0$ and some intermediate working including $4ag$ term.

A1

Or

Use of PE at top and KE at B

Correct PE and KE

Correct deduction including inequality

(M1)

(A1)

(A1)

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3

- (b) (i) C of Energy

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mga(1 + \sin\theta)$$

M1: Equation for conservation of energy with two KE terms and one or two PE terms including a $\sin\theta$. May see m or 0.3 .

A1: Correct equation.

M1A1

$$v^2 = \left(\sqrt{\frac{9}{2}ag} \right)^2 - 2ga(1 + \sin\theta)$$

$$= \frac{5}{2}ag - 2ag \sin\theta$$

Resolve radially

$$\pm R = -mg \sin \theta + \frac{mv^2}{a}$$

M1: Three term equation from resolving radially.
Correct three terms, but condone signs and replacement of sin by cos.

A1: Correct equation. May see m or 0.3.

M1A1

$$= -mg \sin \theta + \frac{5}{2}mg - 2mg \sin \theta$$

$$= -3mg \sin \theta + \frac{5}{2}mg$$

$$= \left(\frac{3}{4} - \frac{9}{10} \sin \theta \right) g$$

OE (must include g)

A1: Simplified correct final answer.

Condone $\left(\frac{9}{10} \sin \theta - \frac{3}{4} \right) g$

A1

5

(ii) When this reaction is zero,

$$\left(\frac{3}{4} - \frac{9}{10} \sin \theta \right) g = 0$$

M1: Putting their reaction equal to zero.

M1

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$$\sin \theta = \frac{5}{6}$$

θ is 56.4° above horizontal

A1: Correct angle. Accept AWRT 56.44.

A1

2

[10]

Q8.

(a) Using conservation of energy:

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 - mgh$$

M1 for 3 terms, 2 KE and PE

or 4 terms, 2 KE and 2 PE

M1A1

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 - mga(1 - \cos \theta)$$

M1A1 for finding h

M1A1

$$v^2 = u^2 + 2ga(1 - \cos \theta)$$

$$v = (u^2 + 2ga[1 - \cos \theta])^{\frac{1}{2}}$$

AG

A1

5

- (b) Using $F = ma$ radially,

$$mg \cos \theta - N = \frac{mv^2}{a}$$

M1 Correct 3 terms

A1 Correct signs ($-N$ or $+N$)

M1A1

Particle leaves surface of hemisphere when $N = 0$

B1

$$mg \cos \theta = \frac{m}{a}(u^2 + 2ga[1 - \cos \theta])$$

M1

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$$\cos \theta = \frac{u^2}{ga} + 2 - 2 \cos \theta$$

$$\cos \theta = \frac{1}{3} \left(\frac{u^2}{ga} + 2 \right)$$

A1

5

[10]

Q9.

- (a) Using conservation of energy:

$$\frac{1}{2}mv^2 = 3mg(1 - \cos \theta)$$

$$M1 \quad \frac{1}{2}mv^2 = mgh$$

M1A1

$$v^2 = 8g(1 - \cos 15)$$

m1

$$v = (8g[1 - \cos 15])^{\frac{1}{2}} \\ = 1.42$$

SC3: 1.41

A1

4

- (b) When particle is at rest,
resolve radially $T = mg \cos 15$

$$M1 \quad T - mg \cos 15 = \frac{mv^2}{r} \text{ or } T = mg \sin 15$$

M1A1

$$22 = mg \cos 15 \\ m = 2.32$$

A1

3

[7]

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Q10.

(a) $\frac{1}{2}mv^2 = \frac{1}{2}m \times 8^2 - mg^2$

M1 3 terms, 2 KE and 1 PE

M1A1

$$v^2 = 64 - 39.2 \\ = 24.8 \\ v = 4.98$$

Accept $\sqrt{24.8}$

A1

3

- (b) Using $F = ma$ radially:

$$R = mg \cos 60 + \frac{mv^2}{r}$$

M1 3 correct terms (not necessarily correct signs)

M1 A1

B1 for 60°

B1

$$= 6g \cos 60 + \frac{6 \times 24.8}{4}$$

$$= 66.6 \text{ N}$$

A1

4

[7]

Q11.

- (a) By conservation of energy to point where QP makes an angle θ with upward vertical:

$$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 - mga(1 + \sin \theta)$$

for 3 terms, 2 KE and 1 PE

mga(1 + sin θ) term

M1A1

$$v^2 = u^2 - 2ag(1 + \sin \theta)$$

A1

Resolve radially

$$R = \frac{mv^2}{a} - mg \sin \theta$$

M1 for 3 terms, include sin θ or cos θ

M1A1

$$= \frac{mu^2}{a} - 3mg \sin \theta - 2mg$$

AG

A1

6

- (b) When particle leaves the track, $R = 0$

M1

$$0 = 3mg - 3mg \sin \theta - 2mg$$

A1

$$\sin \theta = \frac{1}{3}$$

$$SC3 \sin^{-1} \frac{1}{3}$$

M1

$$\theta = 19.5^\circ$$

accept 19.4° or $\theta = 0.340^\circ$

A1

4

[10]

Q12.

(a) Conservation of energy:

$$\frac{1}{2} m (3\sqrt{ag})^2 + mg2a = \frac{1}{2} mv^2$$

M1 for 3 terms: 2 KE and PE

M1A1

$$\frac{9}{2} mga + 2mga = \frac{1}{2} mv^2$$

A1

$$v = \sqrt{13ag}$$

A1

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4

(b) At A, consider vertical forces:

$$T - mg = \frac{mv^2}{a}$$

M1 for 3 terms, 2 correct

M1A1

$$T = mg + 13mg$$

m1

$$T = 14mg$$

ft from (a)

A1ft

4

[8]

Q13.

- (a) Conservation of energy from B to A:

$$\frac{1}{2}mu^2 + mga = \frac{1}{2}mv^2$$

3 terms, 2 KE and PE

M1A1

$$v^2 = 2ag + u^2$$

A1

$$= 27ag$$

$$v = \sqrt{27ag}$$

A1

4

- (b) At A, speed of particle is $\sqrt{27ag}$
Resolving vertically at A:

$$T = \frac{mv^2}{a} + mg$$

Correct 3 terms

Signs all correct

M1

A1F

$$T = m \frac{28ag}{a}$$

$$T = 28mg$$

ft from (a)

A1F

3

[7]

Q14.

- (a) At top, for complete revolutions:

$$\frac{mv^2}{a}$$

= mg where v is speed at top

M1

$$\therefore v^2 = ag$$

A1

Conservation of energy from B to top:

$$\frac{1}{2}mv^2 + mg2a = \frac{1}{2}mu^2$$

3 terms, 2 KE and PE

M1A1

$$u^2 = 4ag + v^2$$

$$= 5ag$$

$$u = \sqrt{5ag}$$

AG

A1

5

- (b) At C, speed of particle is $\sqrt{3ag}$

B1

Resolving horizontally at C:

$$T = \frac{mv^2}{a}$$

Needs 2 correct terms

M1

$$T = m \frac{3ag}{a}$$

$$T = 3mg$$

A1

3

- (c) No air resistance
Bead is a particle

B1

1

[9]

Q15.

(a) $mg \quad 2a = \frac{1}{2}mv^2$

Energy equation

M1A1

$$v = 2\sqrt{ga}$$

A1

3

(b) $T - mg = \frac{mv^2}{2a}$

All terms for M1, no component

M1A1

$$T = 3mg$$

ft if $T > 0$

A1F

3

[6]

Q16.

- (a) Using conservation of energy (lowest and highest points):

M1

$$\frac{1}{2}m(7v)^2 = \frac{1}{2}mv^2 + 2mga$$

A1 for 7v and v

A1A1

$$\frac{48}{2}v^2 = 2ga$$

Needs 48 or 24

M1

$$\therefore v = \sqrt{\frac{ag}{12}}$$

AG

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A1

5

- (b) Velocity at A is $\sqrt{\frac{ag}{12}}$
Resolving vertically at A:
3 terms

M1

$$m \frac{v^2}{a} + R = mg$$

A1 correct 3 terms, A1 correct signs

A1, A1

$$R = mg - \frac{m}{a} \times \frac{ag}{12}$$

$$\left(1 - \frac{1}{12}\right)mg$$

M1A2

$$= \frac{11}{12}mg$$

Condone $-\frac{11}{12}mg$

A1

4

[9]

Q17.

(a) $\frac{1}{2}mv^2 = \frac{1}{2}m \times 2^2 + mg(3 - 3\cos\theta)$

Three term energy equation

Correct equation

M1

A1

$$v^2 = 4 + 6g(1 - \cos\theta)$$

AG
Solving for v^2 .

dM1

4

Correct result from correct working

A1

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(b) $mg \cos\theta = m \frac{v^2}{3}$

Resolving towards the centre

M1

Correct equation

A1

$$3g \cos\theta = 4 + 6g - 6g \cos\theta$$

Solving for $\cos\theta$

dM1

$$\cos\theta = \frac{4 + 6g}{9g}$$

Correct $\cos\theta$

A1

$$\theta = 44.6^\circ$$

Correct angle

A1

5

[9]

Q18.

(a) $\frac{1}{2} mU^2 = \frac{1}{2} mv^2 + mgl (1 - \cos 60^\circ)$

three / four term energy equation with a trig term

M1

$$U^2 = v^2 + gl$$

correct equation

A1

$$v = \sqrt{U^2 - gl}$$

solving for v or v^2

dM1

correct v in a simplified form

A1

4

(b) $T - mg \cos 60^\circ = m \frac{v^2}{l}$

resolving towards the centre of the circle with three terms

M1

$$T = m \left(\frac{U^2 - gl}{l} + \frac{g}{2} \right) = m \left(\frac{U^2}{l} - \frac{g}{2} \right)$$

dM1

substituting for v^2
correct equation

A1

making T the subject

dM1

correct expression for T.
Simplification not necessary.

A1

5

[9]

Q19.

(a) $\frac{1}{2} mU^2 = \frac{1}{2} mv^2 + mgl(1 - \cos 60^\circ)$
three/four term energy equation with a trig term

M1

$U^2 = v^2 + gl$
correct equation

A1

solving for v or v^2

dM1

$V = \sqrt{U^2 - gl}$
correct v in a simplified form

A1

4

(b) $T - mg \cos 60^\circ = m \frac{v^2}{l}$
resolving towards the centre of the circle
with three terms

M1

$T = m \left(\frac{U^2 - gl}{l} + \frac{g}{2} \right) = m \left(\frac{U^2}{l} - \frac{g}{2} \right)$
substituting for v^2

dM1

correct equation

A1

making T the subject

dM1

correct expression for T.
Simplification not necessary.

A1

5

(c) $T - mg = m \frac{U^2}{l}$

considering the vertical forces and using Newton's

second law with $\frac{U^2}{l}$

M1

$$T = m \left(\frac{U^2}{l} + g \right)$$

correct T

A1

2

[11]

Q20.

(a) $\frac{1}{2} mU^2 = mga$

Conservation of energy for M1

$v = 0$ for A1

M1A1

$$U = \sqrt{2ga}$$

ft slip (eg $h = 2a$)

A1F

3

(b)



$$R = 0: mg \cos \theta = \frac{mv^2}{a}$$

M1 for $F = ma$ in general position

M1A1

$$v^2 = ag \times \frac{h}{a}$$

Subs for $\cos \theta$

m1

$$v^2 = hg$$

ft errors in height

A1F

$$\frac{1}{2}m\left(\frac{5ag}{2}\right) = \frac{1}{2}mv^2 + mgh$$

M1 conservation of energy using u, v and h.

M1A1

$$\frac{5ag}{2} = 3gh$$

Subs for v²

m1

$$h = \frac{5a}{6}$$

A1

8

[11]

Q21.

- (a) KE at Q = Change in PE from P to Q
Any one term considered

B1

Attempt at eqn – KE and PE included

M1

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$$\frac{1}{2}mv^2 = mgr(\cos 30^\circ - \cos \theta)$$

Fully correct

A1

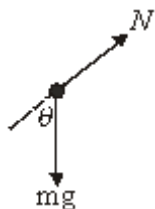
$$v^2 = gr(\sqrt{3} - 2\cos \theta)$$

AG

A1

4

(b)



$$\frac{mv^2}{r} \text{ used}$$

$$mg \cos \theta - N = \frac{mv^2}{r}$$

$$\text{Res force} = \frac{mv^2}{r} \text{ for M1}$$

M1A1B1

$$mg \cos \theta - N = mg (\sqrt{3} - 2 \cos \theta)$$

–use of v^2 from (a)

M1

$$N = mg (3 \cos \theta - \sqrt{3})$$

Must rearrange for $N = \dots$

A1

5

(c) When $\theta = \alpha$, $N = 0$

$$\therefore 3 \cos \alpha - \sqrt{3} = 0$$

$N = 0$ and solve

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M1

$$\cos \alpha = \frac{\sqrt{3}}{3}$$

$$\alpha = 55^\circ$$

Follow through but $30^\circ < \alpha < 90^\circ$

A1ft

2

[11]

Q22.

(a) use of mgh

M1

$$60(9.8)(20 - 20 \cos 60^\circ)$$

$20 \cos 60^\circ$ seen

B1

= 5880 J

A1

3

(b) (i) energy at M = energy at P

$$\frac{1}{2} mu^2 + mgh = \frac{1}{2} mv^2$$

$$\frac{1}{2} (60)(6)^2 + 60(9.8)(20 - 20 \cos \theta)$$

M1 one term correct

A1 two terms correct

M1

A1

$$= \frac{1}{2} (60)v^2$$

fully correct

A1

$$18 + 196 - 196 \cos \theta = \frac{1}{2} v^2$$

$$v^2 = 428 - 392 \cos \theta$$

AG

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A1

4

$$(ii) \quad mg \cos \theta - N = \frac{mv^2}{r}$$

$$\frac{mv^2}{r} \quad \text{used or seen}$$

B1

attempt at Newton's law

M1

$$60(9.8) \cos \theta - N = \frac{60}{20} (428 - 392 \cos \theta)$$

substitute v^2

A1

$$N = 588 \cos \theta - 1284 + 1176 \cos \theta$$

$$\text{or } 1764 \cos \theta - 1284$$

can be unsimplified – must be $N = \dots$

A1

4

(iii) $N > 0$ from M to N

$$1764 \cos \theta - 1284 > 0$$

solve equation or inequality

M1

$$\therefore \cos \theta > \frac{1284}{1764} \approx 0.72789$$

$$\theta < 43.2\dots$$

finding an angle $< 90^\circ$

A1ft

loses contact since $43^\circ < 60^\circ$

correct interpretation

A1

3

Alternative method:

when $\theta = 60^\circ$

evaluate N at end of bridge

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(M1)

$$N = -402 < 0$$

Their N evaluated

(A1ft)

loses contact before end of bridge

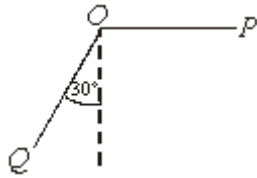
correct interpretation

(A1)

[14]

Q23.

(a)



- (i) At P, $PE = 70g \cos 30^\circ$ $KE = 0$
PE or KE term seen correct (Non zero)

B1

At Q, $KE = \frac{1}{2}(70)v^2$ $PE = 0$

Conservation of energy

$$\frac{1}{2}(70)v^2 = 70g5 \cos 30^\circ$$

Forming equation

M1A1

$$v^2 = 10g \cos 30^\circ$$

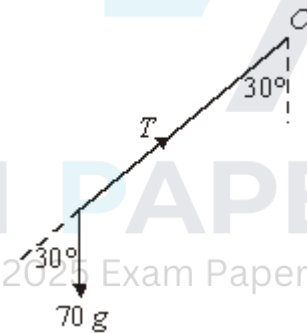
$$\Rightarrow v = 9.21 \text{ ms}^{-1}$$

AG

A1

4

- (ii)



$$\text{Force towards } O = T - 70g \cos 30^\circ$$

For circular motion, $F = ma$

$$\pm(T - 70g \cos 30^\circ) \text{ or } \frac{mv^2}{r} \text{ evaluated}$$

B1

$$\Rightarrow T - 70g \cos 30^\circ = \frac{mv^2}{r}$$

$$\text{For equation - Res. force} = \frac{mv^2}{r} \text{ evaluated}$$

M1A1

5

$$\Rightarrow T - 70g \cos 30^\circ = \frac{70(9.21)^2}{5}$$

$$\Rightarrow T \approx 1782.28 \dots$$

$$\Rightarrow T = 1780 \text{ N}$$

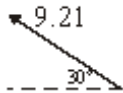
Dependent – substitute and rearrange

m1

AWRT 1780

A1

(b)



$$v^2 - u^2 = 2 \text{ as seen}$$

M1

Vertically, $u = 9.21 \sin 30^\circ$

Initial vertical component seen = $9.21 \sin 30^\circ$

A1

$$v = 0$$

$$a = -9.8$$

$$s = ?$$

Using $v^2 - u^2 = 2as$:

$$\left(\frac{9.21 \sin 30^\circ}{2(9.8)} \right) = 5$$

Substitute values into

$$v^2 - u^2 = 2as$$

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M1

$$s \approx 1.08$$

Approx 1 metre

Must see 1. ...

A1

4

- (c) Height of a man significant to length if rope/distances involved.
Air resistance would reduce speed / height.

Comment that indicates effect of assumption

B1

1

[14]

Q24.

(a) Use of PE = KE

M1

$$mg(5a) = \frac{1}{2}mv^2$$

$$v = \sqrt{10ga}$$

$$\text{or } \sqrt{2g} \sqrt{5a}$$

A1

2

Alternative for (a)

Use of $v^2 = u^2 + 2as$

M1

$$v^2 = 0^2 + 2g(5a)$$

A1

$$v = \sqrt{10ga}$$

(b) (i) Use of energy

$$mg5a - \frac{1}{2}mv^2 + mg2d$$

KE or PE correct

B1

Equation formed (two terms)

M1

$$10ga = v^2 + 4gd$$

Fully correct equation

A1

$$v^2 = 2g(5a - 2d)$$

AG

A1

4

(ii) At Q

$$T + mg = \frac{mv^2}{d}$$

Attempt to use $\frac{mv^2}{r}$ (anything for F)

M1

$$T + mg = \frac{mv^2}{r} \text{ correct (r or d seen)}$$

A1

Using (b)(i) $T + mg = \frac{m2g}{d} (5a - 2d)$
Substitute expression for v^2

A1f.t.

$$T = \frac{2mg}{d} (5a - 2d) - mg$$

$$(or \frac{10mga}{d} - 5mg \text{ OE})$$

A1

4

(iii) For complete vertical circle $T > 0$ at Q

$$\therefore \frac{10mga}{d} - 5mg > 0$$

Sets $T > 0$ or $T = 0$

M1

$$d < 2a$$

AG correctly obtained
(if $T = 0$, must be fully justified)

A1

2

(c) Ball is assumed to be a particle/No air resistance/No jolt at P etc

Any valid reason

B1

1

[13]

Q25.

(a) PE at A = KE at B

$$\frac{1}{2} mu^2 = mgr$$

Attempt to use conservation of energy

M1

$$\Rightarrow u^2 = 2gr$$

A1

2

Alternative

Use $v^2 - u^2 = 2as$

(M1)

$v^2 - 0^2 = 2gr$

$u^2 = 2gr$

(A1)

- (b) (i) Conservation of energy

Energy at B = Energy at P

$mu^2 + mgr \sin \theta = \frac{1}{2} mv^2$

Attempt to use conservation of energy

M1

A1 – one term correct

A1

A1 – all correct

A1

3

$v^2 = 2gr(1 + \sin \theta)$

Use of $u^2 = 2gr$

Printed answer

(A1)

Alternative

Energy at A = Energy at P

$mgr + mgr \sin \theta = \frac{1}{2} mv^2$

Attempt to use conservation of energy

(M1)

A1 – one term correct

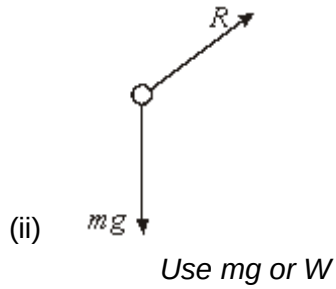
(A1)

A1 – all correct

(A1)

Printed answer

(A1)



B1

1

- (iii) Res force to centre
 $= R - mg \sin \theta$
Seen in equation or stated

B1

$F = ma$ radially gives,

$$R - mg \sin \theta = \frac{mv^2}{r}$$

Follow through error above

M1A1f.t.

$$R = 2mg(1 + \sin \theta) + mg \sin \theta \text{ or } mg(2 + 3\sin \theta)$$

Use of v^2 to find R , need not be simplified

A1

4

(c) $\theta = 90^\circ$, $R = 5mg$

*Either their R with $\theta = 90^\circ$
 or their $R - mg$ without $\theta = 90^\circ$*

M1

Reduces to mg after C since no longer circular motion.

$\therefore 4mg$ reduction
Fully explained for A1

A1

2

(d) $u^2 = 4gr$
Use of (b)(i) to find u^2

B1

Use $v^2 - u^2 = 2as$

$$0^2 - 4gr = 2a(4r)$$

Suitable equation

M1

$$\text{acc} = \frac{-g}{2}$$

$$\text{retardation} = \frac{g}{2}$$

Condone +/-

A1

3

[16]

Q26.

- (a) At C, $N = 0$, so only force on the car will be mg

Stated or implied

B1

Newton's law radially: $mg = \frac{mv^2}{r}$

if "cos θ " used then must state value of $\theta + c$

M1

$$v^2 = gr$$

AG

A1

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3

- (b) Using conservation of energy between B and C:

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + 2mgr$$

Any KE/Pe term seen

M1 equations formed, clear indication of position being considered

B1 M1A1

$$u^2 = 5gr$$

Substitute v^2 and rearrange

A1

4

- (c) Using conservation of energy between A and B:

$$mgh = \frac{1}{2}mv^2$$

$PE = KE$

M1

$$\Rightarrow mgh = \frac{1}{2}m5gr$$

ft their u^2

A1f.t.

$$h = 2.5r$$

$$\Rightarrow k = 2.5$$

A1

3

(d) Conservative forces/friction

E1

1

[11]

Q27.

(a) $a = 0.6 \times 10^2 = 60 \text{ ms}^{-2}$

Use of $a = r\omega^2$

M1

Correct acceleration

A1

2

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Allow ± 60

(b) $R = 0.05 \times 60 = 3\text{N}$

Finding product of mass and acceleration

M1

Correct R

A1ft

2

Follow through incorrect a

(c) $R - 0.05 \times 9.8 = 0.05 \times 60$

Equation of motion at lowest point

M1

Correct equation

A1

$$R = 3.49 \text{ N}$$

Correct R

Follow through incorrect a

A1ft

3

[7]

Q28.

(a) Conservation of energy:

$$\frac{1}{2} m \left(\frac{v}{5} \right)^2 + mg2r = \frac{1}{2} mv^2$$

M1 A1

$$mg4r = m \frac{24}{25} v^2$$

M1

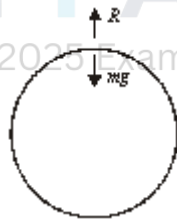
$$v = 5 \sqrt{\frac{gr}{6}}$$

A1

4

(b) (i) At highest point speed is $\sqrt{\frac{1}{6} gr}$

B1



Consider vertical forces:

$$R + \frac{m \left(\sqrt{\frac{1}{6} gr} \right)^2}{r} = mg$$

M1

$$R + \frac{1}{6} mg = mg$$

$$R = \frac{5}{6} mg$$

A1

3

(ii) Using conservation of energy

$$\frac{1}{2} mV^2 = \frac{1}{2} mv^2 - \frac{1}{2} mgr$$

M1

$$= \frac{25}{12} mgr - \frac{1}{2} mgr$$

$$= \frac{19}{12} mgr$$

$$V = \sqrt{\frac{19}{6} gr}$$

Or

$$\frac{1}{2} mV^2 = \frac{1}{2} mv^2 - \frac{1}{2} mgr \quad \text{M1}$$

$$\frac{mV^2}{r} = R - mg \cos 60 \quad \text{M1 A1}$$

Eliminate V

$$mv^2 - mgr = Rr - mgr \cos 60 \quad \text{A1}$$

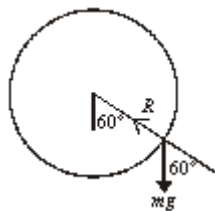
$$\frac{25}{6} mg - mg = R - \frac{1}{2} mg$$

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$$R = \frac{11}{3} mg \quad \text{A1}$$

A1

Resolve radially



$$\frac{mV^2}{r} = R - mg \cos 60$$

M1 A0 if incorrect angle [ie not 60]

M1 A1

$$\frac{19}{6} mg = R - \frac{1}{2} mg$$

$$R = \frac{11}{3} mg$$

A1

5

[12]

Q29.

- (a) Using conservation of energy

$$\frac{1}{2} mv^2 = mga(1 - \cos \theta)$$

M1 A1

$$v^2 = 2ga(1 - \cos \theta)$$

$$v = \sqrt{2ga(1 - \cos \theta)}$$

A1

3

- (b) Particle leaves the surface when

$$mg \cos \theta = \frac{mv^2}{a}$$

M1 A1

$$mg \cos \theta = \frac{m}{a} \times 2ga(1 - \cos \theta)$$

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M1

$$\cos \theta = 2 - 2 \cos \theta$$

$$\cos \theta = \frac{2}{3}$$

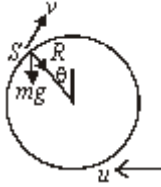
A1

4

[7]

Q30.

Let velocity at point S be v



- (a) Conservation of energy

$$\frac{1}{2}mv^2 + mga(1 + \cos\theta) = \frac{1}{2}mu^2$$

M1 A1

$$v^2 = u^2 - 2ga(1 + \cos\theta)$$

$F = ma$ radially;

$$R + mg \cos\theta = \frac{mv^2}{a}$$

M1 A1

$$R = \frac{mu^2}{a} - 2mg(1 + \cos\theta) - mg \cos\theta$$

M1

$$R = \frac{mu^2}{a} - 3mg \cos\theta - 2mg$$

A1

- (b) For ball to complete the revolution,

$R \geq 0$ at top of track

$$R = \frac{mu^2}{a} - 5mg \geq 0$$

$$u \geq \sqrt{5ag}$$

A1

2

[8]

Q31.

- (a) Speed at lowest point is $4v$

B1

Using C of energy (highest and lowest point)

$$\frac{1}{2}m(4v)^2 - \frac{1}{2}m(v)^2 = mg \cdot 2r$$

M1A1

$$\frac{15}{2}mv^2 = 2mgr$$

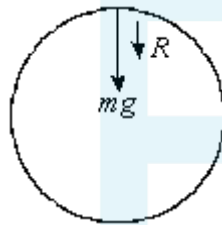
$$v = \sqrt{\frac{4}{15}gr}$$

$$v \text{ and } \frac{v}{4} \text{ used} \Rightarrow \sqrt{\frac{64gr}{15}} \text{ SC2}$$

A1

4

(b) (i)



Using $F = ma$ radially at top:

$$\frac{mv^2}{r} = mg + R$$

M1A1

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$$R = -\frac{mv^2}{r} + mg$$

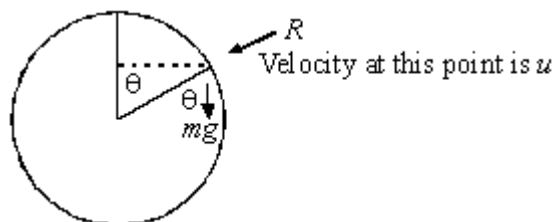
$$= -\frac{4}{15}mg + mg$$

$$\therefore \text{Reaction is } \frac{11}{5}mg$$

A1

3

(ii)



where

$$\frac{1}{2}mu^2 - \frac{1}{2}mv^2 = mgr(1 - \cos\theta)$$

M1

A1

$$(\theta = 60^\circ)$$

B1

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mgr \cdot \frac{1}{2}$$

$$u^2 = \frac{19}{15}gr$$

$$\text{Resolving normally } m \frac{u^2}{r} = mg \cos\theta + R$$

M1

$$R = -mg \cos 60 + \frac{m}{r} \cdot \frac{19}{15}gr$$

$$= \left(-\frac{1}{2} + \frac{19}{15}\right)mg$$

$$\therefore R = \frac{23}{30}mg$$

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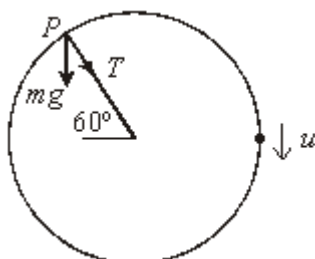
A1

5

[12]

Q32.

(a)



At P when string is inclined at 60°

$$T = 0$$

Resolving along string;

$$\frac{mv^2}{a} = mg \cos 30$$

M1 A1

$$v^2 = ag \frac{\sqrt{3}}{2} \therefore v = \sqrt{\frac{\sqrt{3}}{2} ag}$$

A1

3

(b) Using conservation of energy

$$\frac{1}{2} mu^2 - mga \sin 60 = \frac{1}{2} mv^2$$

M1 A1

$$v^2 = u^2 - 2ag \frac{\sqrt{3}}{2}$$

M1

$$ag \frac{\sqrt{3}}{2} = u^2 - ag \sqrt{3}$$

$$\therefore u^2 = \frac{3\sqrt{3}}{2} ag$$

or

$$u = \sqrt{\frac{3\sqrt{3}}{2} ag}$$

A1

4

[7]

Examiner reports

Q2.

In part (a) most students used two kinetic energy terms and either one or two potential energy terms. The calculation of the difference in height between the two points P and A caused problems to many students. Some students added the potential energy change from P to A to the kinetic energy at P . In part (b) most students tried to include the three

terms T , $mg \cos 25$ and $\frac{mv^2}{a}$ but $\cos 25$ (or $\sin 25$) was sometimes seen as a factor within the wrong term.

Q3.

In part (a) most students considered the kinetic energy at the lowest point and the highest point together with the difference in the potential energy. Unfortunately many forgot that in their kinetic energies, the velocities were $2u$ and $5u$, so the kinetic energies

were $\frac{1}{2}m(2u)^2$ and $\frac{1}{2}m(5u)^2$.

Many incorrectly used $\frac{1}{2}m2u^2$ and $\frac{1}{2}m5u^2$, only inserting the brackets when they found that they had obtained an incorrect answer.

had obtained an incorrect answer. In part (b) a significant number of students did not attempt to find the two equations needed, one considering energy and the other considering forces. For those who did attempt to use these two equations, common errors shown were:

in the conservation of energy equation $\frac{1}{2}m(5u)^2 = \frac{1}{2}mv^2 + mga(1 + \cos 60)$ where the 'height difference' was incorrect or where $\frac{1}{2}mu^2$ was used instead of $\frac{1}{2}m(2u)^2$ or $\frac{1}{2}m(5u)^2$ and in the resolving equation $0 \pm R \pm mg \cos 60 + \frac{mv^2}{r}$, where the components were often incorrect.

in the conservation of energy equation $\frac{1}{2}m(5u)^2 = \frac{1}{2}mv^2 + mga(1 + \cos 60)$ where the 'height difference' was incorrect or where $\frac{1}{2}mu^2$ was used instead of $\frac{1}{2}m(2u)^2$ or $\frac{1}{2}m(5u)^2$ and in the resolving equation $0 \pm R \pm mg \cos 60 + \frac{mv^2}{r}$, where the components were often incorrect.

Q4.

In part (a), most candidates assumed that conservation of energy was required. Unfortunately, some tried to use $v^2 = u^2 + 2as$. In part (b) candidates had difficulty in identifying their ' u ' and their ' v ', often not stating whether they were considering the tension at point A or at point B . Sometimes the force equation at A (or at B) was simply given as $T = mg$, with candidates ignoring any circular motion. The fact that T_A and T_B were stated to be in the ratio of 2 : 5 caused major problems. Some stated that $T_A = 2$ and

$T_B = 5$ - this was not accepted as it could not produce the correct solution - other candidates used $T_A = 2T$ and $T_B = 5T$, which was valid and often led to a successful solution. A number of candidates used $T = 2T$ and $T = 5T$; if they managed to keep track of which of these T s were T_A , T_B or the T used in their ratio, this was condoned. A

significant number of candidates incorrectly used $T_A = \frac{5}{2}$, which usually resulted in a negative value for u^2 . In part (c), those candidates who were successful in part (b) found $v^2 = 5ag$. However many gave the ratio $u : v$ to be $9 : 5$ rather than $3 : \sqrt{5}$.

Q5.

In part (a) most candidates realised that they needed to consider the kinetic energy at the lowest point and the change in potential energy moving from P to the lowest point. Methods used to find the change in height between points were interesting, but often not correct. The answer for the change in height was often truncated and this caused inaccuracy in the answer for the maximum speed. In part (b), many candidates correctly found an equation involving T .

Q6.

In part (a), most candidates used two kinetic energy terms and either one or two potential energy terms. They quickly found that $8v^2 = 2ag$, but some could not solve this to find a value for v .

In part (b), most candidates realised that the maximum and minimum values of the tension were at the lowest and highest positions of the bead, but there were a number of sign errors in the resolving necessary to find the tensions. A significant number of candidates found the maximum

tension to be $\frac{m(5v)^2}{a} = mg$ but did not substitute the value of v which they had found in part (a).

Q7.

In part (a), most candidates considered the kinetic energy at the lowest point and the potential energy at the highest point. Unfortunately many equated these energies and thus obtained

$u = 2\sqrt{ag}$. Noticing the result required, these then just changed their results to $u > 2\sqrt{ag}$; this was not accepted.

In part (b)(i), many candidates attempted to find at least one of the two equations required.

Common errors seen were in the conservation of energy equation, where the 'height' in

$\frac{1}{2} mu^2 = \frac{1}{2} mv^2 + mga(1 + \sin \theta)$ was often incorrect, and in the resolving equation

$\pm R = -mg \sin \theta + \frac{mv^2}{r}$, where the $\sin \theta$ term was often incorrect. Few candidates obtained the correct value for R .

In part (b)(ii), virtually all candidates equated R to zero but rarely solved their resulting equation.

Q8.

In part (a) most candidates used two kinetic energy terms and either one or two potential energy terms. Unfortunately those who used two potential energy terms often started with

$$\frac{1}{2}mv^2 + mga = \frac{1}{2}mu^2 + mga(1 - \cos\theta) \quad \text{instead of} \quad \frac{1}{2}mv^2 + mga \cos\theta = \frac{1}{2}mu^2 + mga$$

Naturally all candidates obtained the printed result!

In part (b) candidates appreciated that the reaction between the hemisphere and the particle was zero when the particle left the surface of the hemisphere. Unfortunately relatively few candidates resolved in the correct direction and the required equation of mg

$$\cos\theta = \frac{\frac{mv^2}{a}}{a} \quad \text{was therefore not found.}$$

Q9.

In part (a), most candidates attempted to use conservation of energy. Frequently accuracy was lost through truncation or premature rounding. This led to common incorrect answers

of 1.4, 1.41 or even $\sqrt{2}$, whereas 1.42 was the required answer. In part (b), most candidates resolved vertically at Q, obtaining $T \cos 15 = mg$. This ignored the transverse acceleration. Hence resolving radially was required, and this produced the correct result of $T = mg \cos 15$.

Q10.

In part (a), most candidates used two kinetic energy terms and one potential energy term.

Unfortunately, signs were frequently incorrect. Again the numerical value of $\sqrt{24.8}$ was sometimes given as 5, rather than 4.98. In part (b), candidates rarely showed a full force

diagram. Thus, when using $F = ma$ radially, they commonly obtained $R = \frac{\frac{mv^2}{r}}{r}$, omitting the term $mg \cos 60$.

Q11.

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Many candidates made little progress in part (a) of this question. A number did not use conservation of energy correctly, with many not finding the change in potential energy to be $mga(1 + \sin\theta)$. It was common for such candidates to resolve radially to obtain $R =$

$\frac{\frac{mv^2}{a} - mg}{a}$, but then use the answer to change this into $R = \frac{\frac{mv^2}{a} - mg \sin\theta}{a}$. The printed answer meant that many candidates changed their original terms, enabling their answers to appear correct. However, they were rarely changed correctly. Part (b) was usually

answered well, but a common error was, when $u = \sqrt{3ag}$, to give $\frac{\frac{mu^2}{a} = \frac{m\sqrt{3}}{a}ag}{a}$ instead of $\frac{m \cdot 3ag}{a}$, which was $3mg$.

Q12.

The correct method was usually seen in part (a), but difficulty occurred in simplifying

$\frac{1}{2}mv^2$, where $v = 3\sqrt{ag}$. Others found some terms as a multiple of g with other terms as

numerical quantities, so instead of $4ag + 9ag = 13ag$, their answer became $39.2a + 9ag$, which was rarely simplified. In part (b), the lack of a clear force diagram often gave

$$T - mg = \frac{mv^2}{a}, \text{ or led to an incorrect sign in } T - mg = \frac{mv^2}{a}.$$

Q13.

Part (a) of this question was answered well.

In part (b), mistakes often arose when $T = \frac{mv^2}{a} + mg$ became $T = \frac{mv^2}{a} - mg$.

Q14.

Many candidates made little progress in this question. A number did not use conservation of energy correctly, assuming that the kinetic energy at the top was zero. The fact that the bead could not complete full revolutions attached to a string with insufficient speed to remain taut at the top was usually ignored. It was common for such candidates to arrive at $v^2 = 4ag$, and then then use many techniques in dubious attempts to justify the addition of the missing ag .

In part (b), the required components, T and $\frac{mv^2}{r}$, appeared frequently in the equation, but often candidates did not find the value of v when the bead was at C. Part (c) was usually answered well.

Q15.

Part (a) was popular and done well. Part (b) was found harder than expected, with some considering the situation as static, and others omitting the weight, while sign errors were quite frequent.

Q16.

Many candidates completed part (a) well. A number did not use conservation of energy

whilst many considered $\frac{1}{2}m(7v)^2$ to be $\frac{7}{2}mv^2$; regardless of these errors it was common

for such candidates to arrive at the printed result $v = \sqrt{\frac{ag}{12}}$. In part (b), the required

components, R , mg and $\frac{mv^2}{r}$, appeared frequently in the equation but often not with the appropriate signs.

Q17.

Many of the candidates were able to obtain the printed answer, but sometimes from very dubious working. There were a number of candidates who tried to use the constant acceleration equation $v^2 = u^2 + 2as$. They did obtain the correct result, but their method was not valid. The candidates that started with a clear statement or equation based on conservation of energy, were usually able to complete the solution correctly. In part (b), there were fewer correct responses than in part (a). Most of the candidates realised that they needed a zero reaction force, writing $R = 0$. The main cause of errors was either not

resolving the weight at all or not resolving the weight correctly. A few candidates obtained correct values for $\cos \theta$ but gave incorrect angles.

Q18.

This question proved to be more challenging and only a relatively small number of candidates gained full marks. Part (a) was done well by those candidates who could form the correct energy equation. Some candidates only included one kinetic energy term.

A number of candidates obtained a correct expression for v^2 , but simplified it incorrectly.

In part (b), there were a variety of responses. For example, some simply assumed that the tension would be the product of the mass and the acceleration.

For those candidates who formed a three term equation of motion, there were a number of problems with resolving, for example using $T \cos 60^\circ$ instead of T . Some students who had formed correct equations were unable to carry out the simplification correctly and so gave an incorrect final answer.

The most common error was to substitute the value of v obtained in part (a) instead of simply substituting U .

Q19.

This question proved to be more challenging and only a relatively small number of candidates gained full marks. Part (a) was done well by those candidates who could form the correct energy equation. Some candidates only included one kinetic energy term. A number of candidates obtained a correct expression for v^2 , but simplified it incorrectly.

In part (b), there were a variety of responses. For example, some simply assumed that the tension would be the product of the mass and the acceleration.

For those candidates who formed a three term equation of motion, there were a number of problems with resolving, for example using $T \cos 60^\circ$ instead of T . Some students who had formed correct equations were unable to carry out the simplification correctly and so gave an incorrect final answer.

Part (c) was often done well. The most common error was to substitute the value of v obtained in part (a) instead of simply substituting U .

Q20.

This proved, as expected, to be the most demanding question on the paper. Part (a) was generally successfully completed. To score marks in part (b) it was necessary to consider a general position when applying the energy principle and when constructing an equation of motion. Linking these equations and also linking angles and vertical distances proved beyond many in this unstructured request. Able and well prepared candidates produced excellent solutions which were both concise and accurate in algebraic manipulation.

Q21.

Using energy is well understood and part (a) proved quite successful. A few candidates confused themselves by quoting $r(1\cos - \theta)$ and then failing to apply it correctly to obtain the printed result. Part (b) proved far more difficult and it is still a great concern that candidates struggle with this concept. A couple of errors were common:

- Use of $N\cos\theta$ or $N\sin\theta$ in $F=ma$
- Resultant force in the wrong direction
- Resultant force expressed as a sum rather than a difference

Many candidates realised that they had to solve $N = 0$, but incorrect results in part (b)

meant they had created an impossible equation e.g. $\cos\theta = \sqrt{3}$ or required addition of sine wave methods to solve it.

Q22.

A very pleasing response was made to this question. Formulae for potential energy and kinetic energy are well known. In part (b) candidates were sometimes guilty of double-counting energy terms by muddling up what applied where. Those candidates who considered the situation at the start and at the end were generally more successful. A few candidates still quoted $v^2 = u^2 + 2as$ for part (b)(i) despite previous Examiners' reports highlighting this error in the past – no marks are awarded for this. Responses to part (b)(ii) were far better than in the past although sometimes $N - mg \cos\theta$ was seen or, more worrying, the acceleration was treated as 0. Candidates who tried a vector approach were always unsuccessful. Many correct solutions were seen for part (b)(iii). It was pleasing to see that solving an equation to find when $N = 0$ was equally as popular as finding N when $\theta = 60^\circ$. Only a few candidates tried to set up equations with the velocity involved.

Q23.

Using principles involving energy was well understood and part (a)(i) proved to be very successful. Occasionally some candidates confused themselves by identifying the wrong change in height when calculating potential energy. Part (a)(ii) proved to be far more difficult, many candidates seeming to struggle with this concept. A variety of errors were seen which included: use of $T\cos\theta$ or $T\sin\theta$ in $F = ma$; resultant force in the wrong direction often vertically; and incorrect substitution into mv^2/r . A mixed response was seen to part (b), with a good proportion of candidates realising that the vertical component was required.

Part (c) was more discriminating where candidates who commented on an assumption and considered its likely effect obtained the mark.

Q24.

A very good response was made to this question. Candidates were able to use conservation of energy well in both parts (a) and (b)(i). Explanations were better than in the past with candidates stating the positions considered. A few candidates quoted $v^2 = u^2 + 2as$ for part (b)(i). No marks were awarded for this. Responses to part (b)(ii) were good

with $T + mg = \frac{mv^2}{r}$ correctly used. Many candidates were then able to use their expression in part (iii) to solve an inequality. Many correct assumptions were seen for part (c).

Q25.

Work here showed an improved response in comparison with previous papers, even though the question was non-standard in some parts. Candidates generally found no problem with part (a). It was, however, disappointing to see some candidates use ' $v^2 - u^2 = 2as$ ' in part (b)(i) and, as a result, score nothing.

Many candidates drew a correct force diagram and, as a direct result, many correct answers were obtained in part (b)(iii).

Part (c) was the most demanding part of the question and few candidates gave a fully correct response, although many scored 1 mark for either considering their expression from part (b)(iii) or for finding the normal reaction value when $\theta = 90^\circ$.

Success in part (d) generally depended on how well candidates had followed the story in the question. Most used ' $v^2 - u^2 = 2as$ ' correctly but several correct approaches using work done = change in kinetic energy were also seen.

Q26.

This vertical circle question prompted the best response to any seen so far. Focusing on this topic for coursework does seem to help.

In part (a), some candidates scored zero by failing to consider forces at C. However, most provided excellent solutions.

In parts (b) and (c), marks were sometimes lost when energy terms from different points got mixed up. Candidates who quoted $k = 2.5$ from memory failed to score.

A good range of improvements was suggested in part (d), with the inclusion of air resistance and friction being the most favoured. It was surprising to see that some candidates did not seem to understand the request "to refine".

Q27.

While there were many good solutions to this question, there were also a number of errors that appeared on quite a few scripts.

In part (a) several candidates included the mass when calculating the acceleration.

In part (b) some candidates assumed that the only force was the weight of the sphere.

Finally in part (c), the most common error was to find the difference between the weight and the resultant force, rather than the sum of them.

Q28.

Most candidates were successful in completing part (a). However, a number of

candidates used v as the velocity at the lowest point, obtained $v = \sqrt{\frac{gr}{6}}$ and did not appreciate the way to obtain the printed result.

In part (b) a significant proportion of candidates used v to be $5\sqrt{\frac{gr}{6}}$ at all of the positions under consideration. Apart from this, the common error by candidates was to use incorrect signs in their use of either centripetal acceleration or centrifugal force.

Q29.

Most candidates answered this question well, although the weakest did fudge the algebra to obtain the printed result. Few candidates drew a force diagram in part (b), and these relied upon intuition to obtain the required result.

Q30.

This question was answered well. Only a few candidates had to fiddle the algebra to obtain the printed results.

Q31.

This question was completed well. Errors occurred in the direction of the forces; in particular centripetal acceleration caused major problems with directions.

Q32.

This question was answered well by the majority who appreciated that part (a) could be solved as a result of resolving forces at the instant when the string became slack, and part (b) was from the use of conservation of energy. There were a few arithmetical errors in part (b).



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